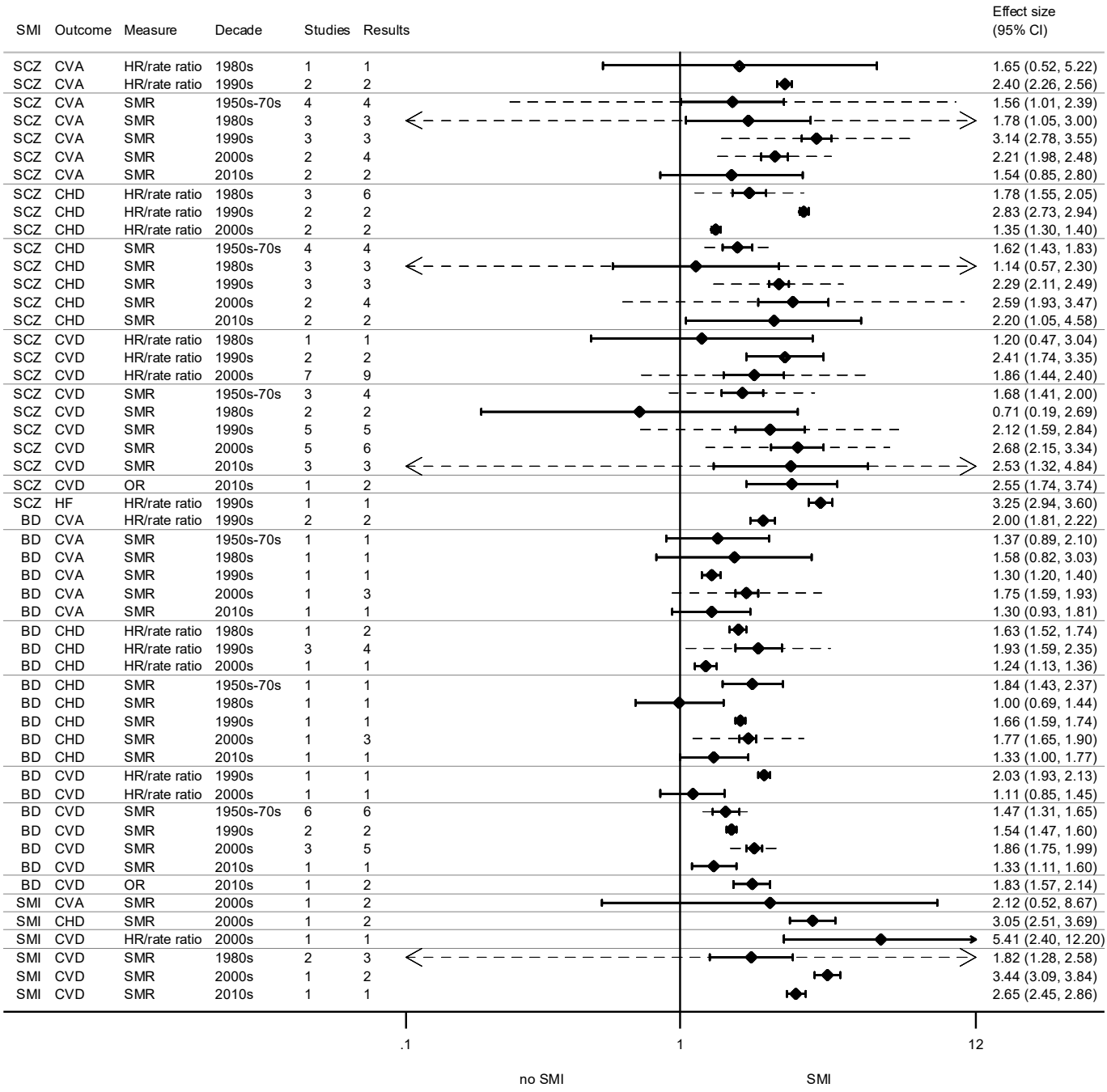
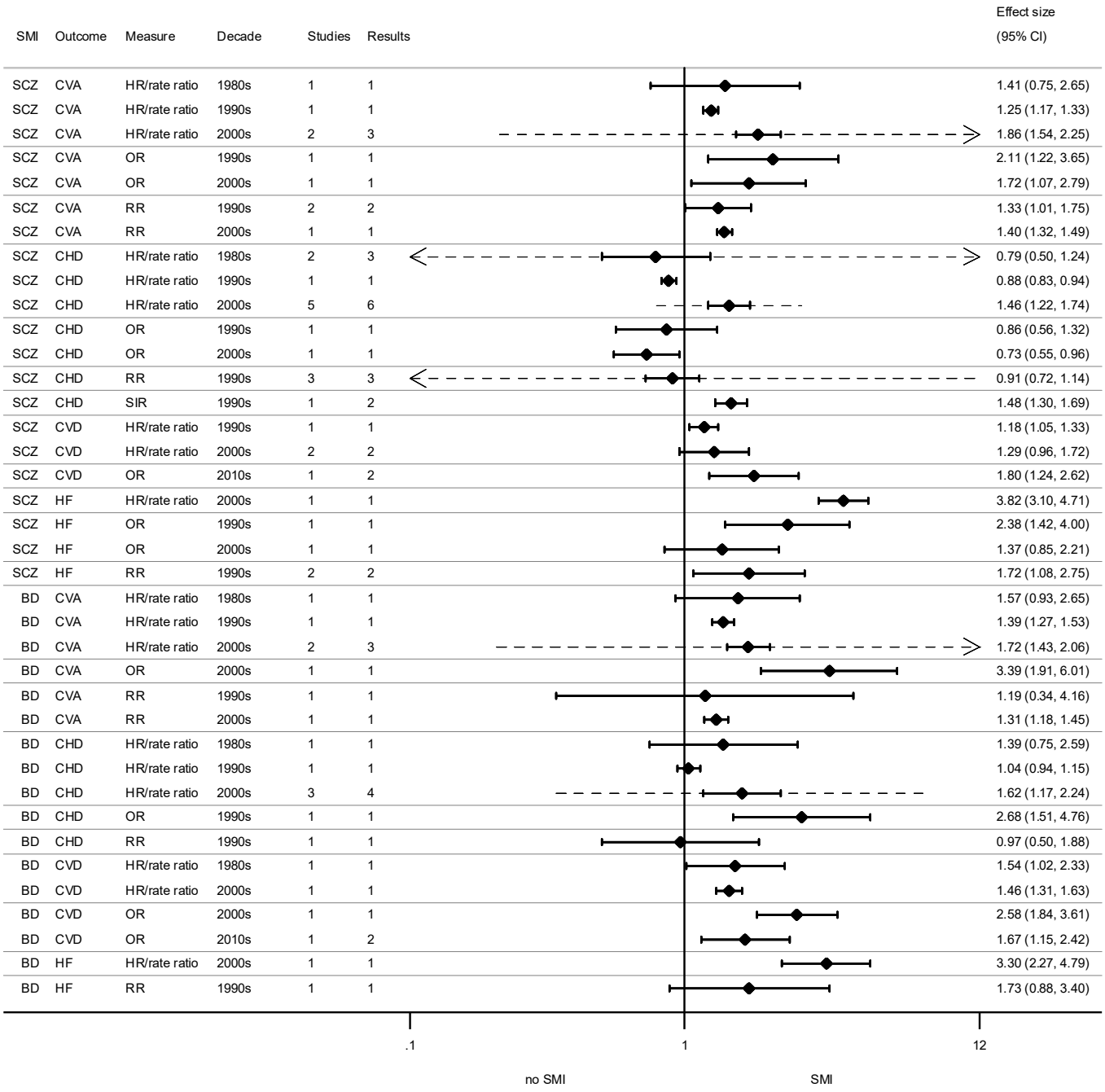
S19 File. Analysis of temporal trends



SCZ – schizophrenia, BD – bipolar disorder, SMI – mixed schizophrenia and bipolar disorder, CHD – coronary heart disease, CVA – cerebrovascular accident, CVD – all circulatory disease, HR – hazard ratio, OR – odds ratio, SMR – standardised mortality ratio

Pooled estimates with 95% confidence intervals (solid lines) and 95% prediction intervals if at least 3 results (dashed lines)

Fig A: Forest plot showing pooled estimates of cardiovascular mortality ratios by severe mental illness, outcome and decade



SCZ – schizophrenia, BD – bipolar disorder, CHD – coronary heart disease, CVA – cerebrovascular accident, CVD – major cardiovascular events, HF – heart failure, HR – hazard ratio, OR – odds ratio, RR – risk ratio, SIR – standardised incidence ratio

Pooled estimates with 95% confidence intervals (solid lines) and 95% prediction intervals if at least 3 results (dashed lines)

Fig B: Forest plot showing pooled estimates of cardiovascular incidence ratios by severe mental illness, outcome and decade

Table A: Meta-analysis: pooled results by severe mental illness and cardiovascular mortality

SMI	Outcome	Measure	Decade	Studies	Results	Effect	95% CI			p	Lower prediction interval	Upper prediction interval	I^2	Q	Q df
SCZ	CVA	HR/RRatio	1980s	1	1	1.65	(0.52	-	5.22)	0.394	.	.	.	0.0	0
SCZ	CVA	HR/RRatio	1990s	2	2	**2.40**	(2.26	-	2.56)	<0.001	.	.	0.0	0.1	1
SCZ	CVA	SMR	1950-70s	4	4	**1.56**	(1.01	-	2.40)	0.045	0.2	10.1	74.2	11.6	3
SCZ	CVA	SMR	1980s	3	3	**1.78**	(1.05	-	3.00)	0.032	0.0	620.7	66.1	5.9	2
SCZ	CVA	SMR	1990s	3	3	**3.14**	(2.78	-	3.55)	<0.001	1.4	6.9	0.0	0.6	2
SCZ	CVA	SMR	2000s	2	4	**2.21**	(1.98	-	2.48)	<0.001	1.4	3.6	74.4	11.7	3
SCZ	CVA	SMR	2010s	2	2	1.54	(0.85	-	2.80)	0.158	.	.	95.6	22.5	1
SCZ	CHD	HR/RRatio	1980s	3	6	**1.78**	(1.55	-	2.05)	<0.001	1.1	2.8	90.6	53.1	5
SCZ	CHD	HR/RRatio	1990s	2	2	**2.83**	(2.73	-	2.94)	<0.001	.	.	0.0	0.7	1
SCZ	CHD	HR/RRatio	2000s	2	2	**1.35**	(1.31	-	1.41)	<0.001	.	.	0.0	0.1	1
SCZ	CHD	SMR	1950-70s	4	4	**1.62**	(1.43	-	1.83)	<0.001	1.2	2.1	0.0	1.2	3
SCZ	CHD	SMR	1980s	3	3	1.14	(0.57	-	2.30)	0.710	0.0	6779.8	89.6	19.3	2
SCZ	CHD	SMR	1990s	3	3	**2.29**	(2.11	-	2.49)	<0.001	1.3	3.9	0.0	0.0	2
SCZ	CHD	SMR	2000s	2	4	**2.59**	(1.93	-	3.47)	<0.001	0.6	10.8	98.7	222.9	3
SCZ	CHD	SMR	2010s	2	2	**2.20**	(1.05	-	4.59)	0.036	.	.	98.8	82.3	1
SCZ	CVD	HR/RRatio	1980s	1	1	1.20	(0.47	-	3.04)	0.700	.	.	.	0.0	0
SCZ	CVD	HR/RRatio	1990s	2	2	**2.41**	(1.74	-	3.35)	<0.001	.	.	88.4	8.6	1
SCZ	CVD	HR/RRatio	2000s	7	9	**1.86**	(1.44	-	2.40)	<0.001	0.7	4.8	98.4	512.9	8
SCZ	CVD	OR	2010s	1	2	**2.55**	(1.74	-	3.74)	<0.001	.	.	91.8	12.1	1
SCZ	CVD	SMR	1950-70s	3	4	**1.68**	(1.41	-	2.00)	<0.001	0.9	3.1	39.1	4.9	3
SCZ	CVD	SMR	1980s	2	2	0.71	(0.19	-	2.69)	0.616	.	.	98.1	51.6	1
SCZ	CVD	SMR	1990s	5	5	**2.13**	(1.59	-	2.84)	<0.001	0.7	6.3	95.7	93.1	4
SCZ	CVD	SMR	2000s	5	6	**2.68**	(2.15	-	3.34)	<0.001	1.2	5.8	98.6	357.9	5
SCZ	CVD	SMR	2010s	3	3	**2.53**	(1.32	-	4.84)	0.005	0.0	11000.0	99.4	350.0	2
SCZ	HF	HR/RRatio	1990s	1	1	**3.25**	(2.94	-	3.60)	<0.001	.	.	.	0.0	0
BD	CVA	HR/RRatio	1990s	2	2	**2.01**	(1.81	-	2.22)	<0.001	.	.	0.0	0.5	1
BD	CVA	SMR	1950s-70s	1	1	1.37	(0.89	-	2.10)	0.151	.	.	.	0.0	0
BD	CVA	SMR	1980s	1	1	1.58	(0.82	-	3.03)	0.172	.	.	.	0.0	0
BD	CVA	SMR	1990s	1	1	**1.30**	(1.20	-	1.40)	<0.001	.	.	.	0.0	0
BD	CVA	SMR	2000s	1	3	**1.75**	(1.59	-	1.93)	<0.001	0.9	3.3	0.0	0.5	2
BD	CVA	SMR	2010s	1	1	1.30	(0.94	-	1.81)	0.119	.	.	.	0.0	0
BD	CHD	HR/RRatio	1980s	1	2	**1.63**	(1.52	-	1.74)	<0.001	.	.	0.0	0.6	1
BD	CHD	HR/RRatio	1990s	3	4	**1.93**	(1.59	-	2.35)	<0.001	1.1	3.5	7.7	3.3	3
BD	CHD	HR/RRatio	2000s	1	1	**1.24**	(1.13	-	1.36)	<0.001	.	.	.	0.0	0
BD	CHD	SMR	1950s-70s	1	1	**1.84**	(1.43	-	2.37)	<0.001	.	.	.	0.0	0
BD	CHD	SMR	1980s	1	1	1.00	(0.69	-	1.44)	0.980	.	.	.	0.0	0
BD	CHD	SMR	1990s	1	1	**1.66**	(1.59	-	1.74)	<0.001	.	.	.	0.0	0

SMI	Outcome	Measure	Decade	Studies	Results	Effect	95% CI			p	Lower prediction interval	Upper prediction interval	I^2	Q	Q df
BD	CHD	SMR	2000s	1	3	**1.77**	(1.65	-	1.90)	<0.001	1.1	2.8	0.0	1.2	2
BD	CHD	SMR	2010s	1	1	**1.33**	(1.00	-	1.77)	0.052	.	.	.	0.0	0
BD	CVD	HR/RRatio	1990s	1	1	**2.03**	(1.93	-	2.13)	<0.001	.	.	.	0.0	0
BD	CVD	HR/RRatio	2000s	1	1	1.11	(0.85	-	1.46)	0.450	.	.	.	0.0	0
BD	CVD	OR	2010s	1	2	**1.83**	(1.57	-	2.14)	<0.001	.	.	65.2	2.9	1
BD	CVD	SMR	1950s-70s	6	6	**1.47**	(1.31	-	1.65)	<0.001	1.2	1.8	7.4	5.4	5
BD	CVD	SMR	1990s	2	2	**1.54**	(1.47	-	1.60)	<0.001	.	.	31.7	1.5	1
BD	CVD	SMR	2000s	3	5	**1.86**	(1.75	-	1.99)	<0.001	1.5	2.3	52.1	8.4	4
BD	CVD	SMR	2010s	1	1	**1.33**	(1.11	-	1.60)	0.002	.	.	.	0.0	0
SMI	CHD	SMR	2000s	1	1	**3.70**	(2.47	-	5.55)	<0.001	.	.	.	0.0	0
SMI	CVA	SMR	2000s	1	1	0.80	(0.16	-	4.08)	0.788	.	.	.	0.0	0
SMI	CVD	HR/RRatio	2000s	1	1	**5.41**	(2.40	-	12.20)	<0.001	.	.	.	0.0	0
SMI	CVD	SMR	1980s	2	3	**1.82**	(1.28	-	2.58)	0.001	0.0	105.6	75.1	8.0	2
SMI	CVD	SMR	2000s	1	1	**3.70**	(2.80	-	4.90)	<0.001	.	.	.	0.0	0
SMI	CVD	SMR	2010s	1	1	**2.65**	(2.45	-	2.86)	<0.001	.	.	.	0.0	0

SCZ – schizophrenia, BD – bipolar disorder, SMI – mixed severe mental illness, HR – hazard ratio, RRatio – rate ratio, SMR – standardised mortality ratio, OR – odds ratio, CVA – cerebrovascular accident, CHD – coronary heart disease, CVD – all circulatory disease, HF – heart failure
Results where 95% confidence intervals exclude the null highlighted in ***bold***

Table B: Meta-analysis: pooled results by severe mental illness and cardiovascular incidence

SMI	Outcome	Measure	Decade	Studies	Results	Effect	95% CI			p	Lower prediction interval	Upper prediction interval	I^2	Q	Q df
SCZ	CVA	HR/RRatio	1980s	1	1	1.41	(0.75	-	2.65)	0.285	.	.	.	0.0	0
SCZ	CVA	HR/RRatio	1990s	1	1	1.25	(1.17	-	1.33)	<0.001	.	.	.	0.0	0
SCZ	CVA	HR/RRatio	2000s	2	3	1.86	(1.54	-	2.25)	<0.001	0.2	16.4	73.3	7.5	2
SCZ	CVA	OR	1990s	1	1	2.11	(1.22	-	3.65)	0.008	.	.	.	0.0	0
SCZ	CVA	OR	2000s	1	1	1.72	(1.07	-	2.79)	0.027	.	.	.	0.0	0
SCZ	CVA	RR	1990s	2	2	1.33	(1.01	-	1.75)	0.040	.	.	0.0	0.2	1
SCZ	CVA	RR	2000s	1	1	1.40	(1.32	-	1.49)	<0.001	.	.	.	0.0	0
SCZ	CHD	HR/RRatio	1980s	2	3	0.79	(0.50	-	1.24)	0.307	0.0	205.5	88.2	17.0	2
SCZ	CHD	HR/RRatio	1990s	1	1	0.88	(0.83	-	0.94)	<0.001	.	.	.	0.0	0
SCZ	CHD	HR/RRatio	2000s	5	6	1.46	(1.22	-	1.74)	<0.001	0.8	2.7	98.0	243.4	5
SCZ	CHD	OR	1990s	1	1	0.86	(0.56	-	1.32)	0.487	.	.	.	0.0	0
SCZ	CHD	OR	2000s	1	1	0.73	(0.55	-	0.96)	0.023	.	.	.	0.0	0
SCZ	CHD	RR	1990s	3	3	0.91	(0.72	-	1.14)	0.403	0.1	11.5	73.7	7.6	2
SCZ	CHD	SIR	1990s	1	2	1.48	(1.30	-	1.69)	<0.001	.	.	85.0	6.7	1
SCZ	CVD	HR/RRatio	1990s	1	1	1.18	(1.05	-	1.33)	0.006	.	.	.	0.0	0
SCZ	CVD	HR/RRatio	2000s	2	2	1.29	(0.96	-	1.72)	0.092	.	.	84.9	6.6	1
SCZ	CVD	OR	2010s	1	2	1.80	(1.24	-	2.62)	0.002	.	.	95.3	21.4	1
SCZ	HF	HR/RRatio	2000s	1	1	3.82	(3.10	-	4.71)	<0.001	.	.	.	0.0	0
SCZ	HF	OR	1990s	1	1	2.38	(1.42	-	4.00)	0.001	.	.	.	0.0	0
SCZ	HF	OR	2000s	1	1	1.37	(0.85	-	2.21)	0.195	.	.	.	0.0	0
SCZ	HF	RR	1990s	2	2	1.72	(1.08	-	2.75)	0.023	.	.	75.1	4.0	1
BD	CVA	HR/RRatio	1980s	1	1	1.57	(0.93	-	2.65)	0.090	.	.	.	0.0	0
BD	CVA	HR/RRatio	1990s	1	1	1.39	(1.27	-	1.53)	<0.001	.	.	.	0.0	0
BD	CVA	HR/RRatio	2000s	2	3	1.72	(1.43	-	2.06)	<0.001	0.2	14.2	76.3	8.4	2
BD	CVA	OR	2000s	1	1	3.39	(1.91	-	6.01)	<0.001	.	.	.	0.0	0
BD	CVA	RR	1990s	1	1	1.19	(0.34	-	4.16)	0.785	.	.	.	0.0	0
BD	CVA	RR	2000s	1	1	1.31	(1.18	-	1.45)	<0.001	.	.	.	0.0	0
BD	CHD	HR/RRatio	1980s	1	1	1.39	(0.75	-	2.59)	0.299	.	.	.	0.0	0
BD	CHD	HR/RRatio	1990s	1	1	1.04	(0.94	-	1.15)	0.425	.	.	.	0.0	0
BD	CHD	HR/RRatio	2000s	3	4	1.62	(1.17	-	2.24)	0.004	0.3	7.7	95.8	70.7	3
BD	CHD	OR	1990s	1	1	2.68	(1.51	-	4.76)	0.001	.	.	.	0.0	0
BD	CHD	RR	1990s	1	1	0.97	(0.50	-	1.88)	0.928	.	.	.	0.0	0
BD	CVD	HR/RRatio	1980s	1	1	1.54	(1.02	-	2.33)	0.040	.	.	.	0.0	0
BD	CVD	HR/RRatio	2000s	1	1	1.46	(1.31	-	1.63)	<0.001	.	.	.	0.0	0

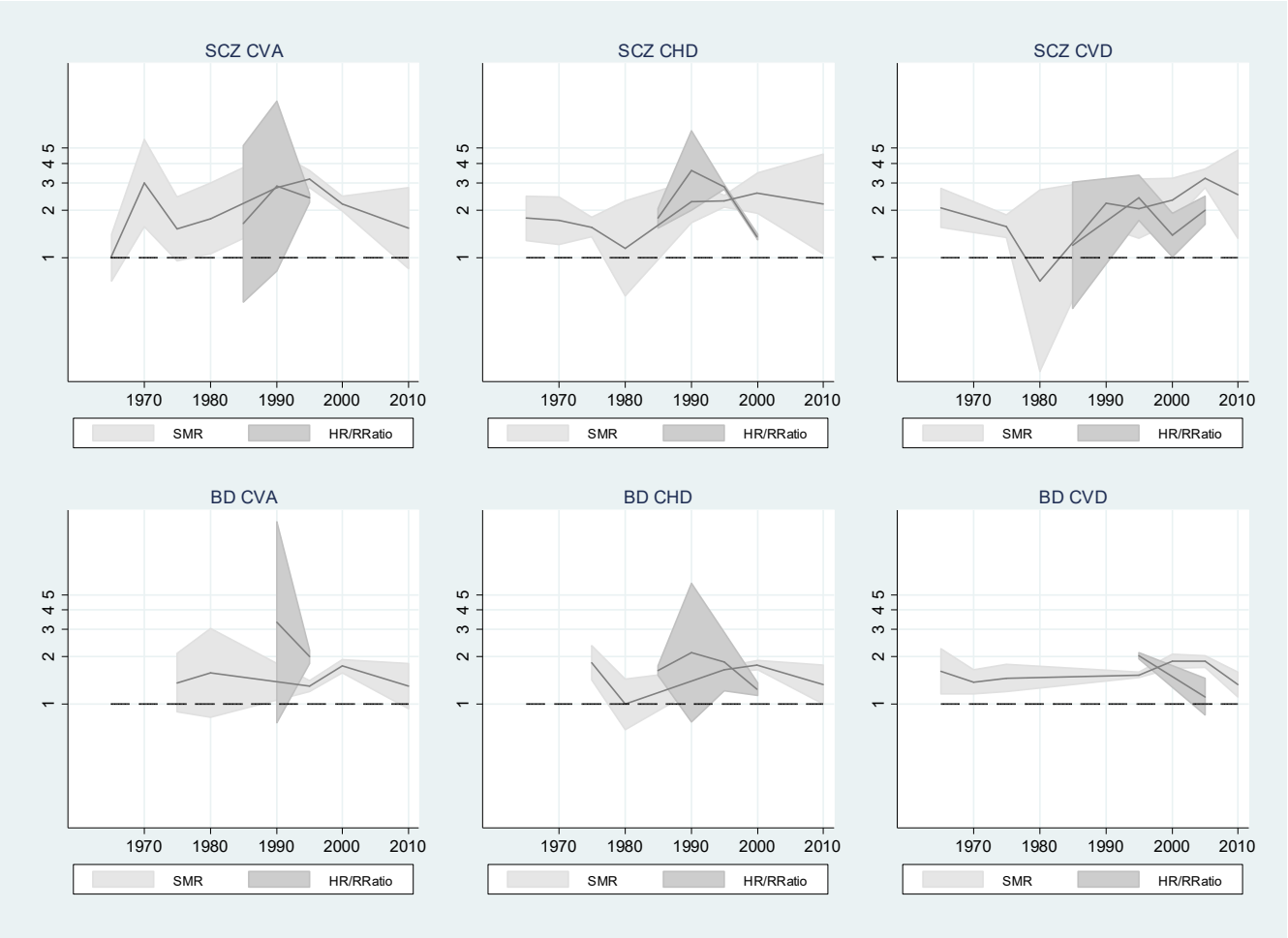
SMI	Outcome	Measure	Decade	Studies	Results	Effect	95% CI	p	Lower prediction interval	Upper prediction interval	I^2	Q	Q df
BD	CVD	OR	2000s	1	1	2.58	(1.84 - 3.61)	<0.001	.	.	.	0.0	0
BD	CVD	OR	2010s	1	2	1.67	(1.15 - 2.42)	0.007	.	.	96.9	31.8	1
BD	HF	HR/RRatio	2000s	1	1	3.30	(2.27 - 4.79)	<0.001	.	.	.	0.0	0
BD	HF	RR	1990s	1	1	1.73	(0.88 - 3.40)	0.111	.	.	.	0.0	0

SCZ – schizophrenia, BD – bipolar disorder, SMI – mixed severe mental illness, HR – hazard ratio, RRatio – rate ratio, OR – odds ratio, RR – risk ratio, SIR – standardised incidence ratio, CVA – cerebrovascular accident, CHD – coronary heart disease, CVD – major cardiovascular events, HF – heart failure
Results where 95% confidence intervals exclude the null highlighted in ***bold***

Table C: Results of meta-regressions: estimates of increased effect size relative to the reference category by median 5-year calendar period of outcome, exponentiated regression coefficients (95% confidence intervals,) p values

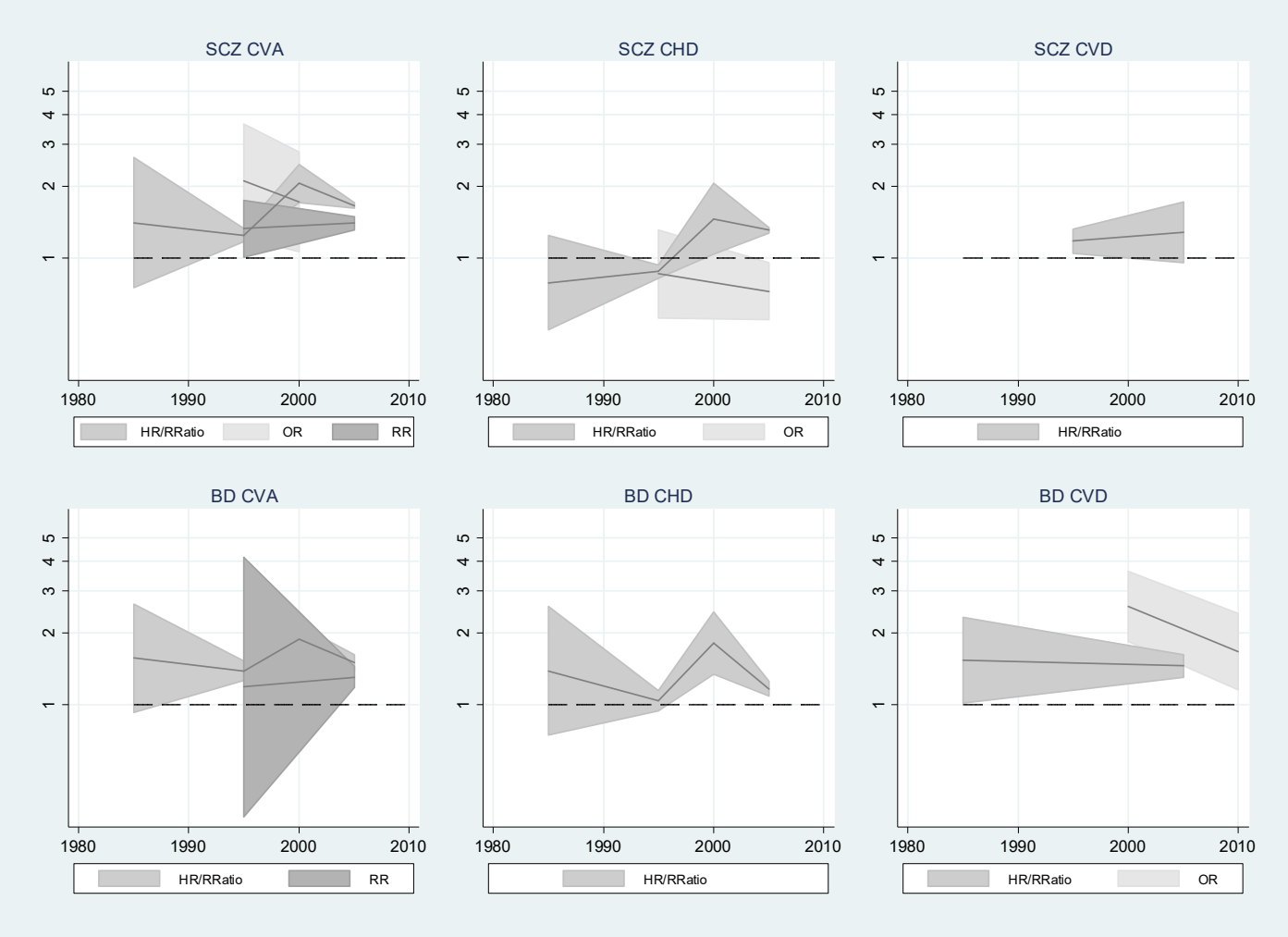
Explanatory variable (if no. of studies/results ≥10)			Schizophrenia			Bipolar disorder
			CVA	CHD	CVD	CVD
5-year calendar period	SMR	1950-69 (ref)	1	1	1	1
		1970-74	3.00 (0.85-10.54), 0.079	0.96 (0.25-3.71), 0.951	-	0.85 (0.55-1.33), 0.430
		1975-79	1.53 (0.60-3.92), 0.328	0.90 (0.29-2.81), 0.830	0.75 (0.20-2.85), 0.643	0.91 (0.60-1.38), 0.611
		1980-84	1.71 (0.69-4.24), 0.208	0.63 (0.21-1.90), 0.360	0.34 (0.08-1.42), 0.127	-
		1985-89	-	-	-	-
		1990-94	2.73 (0.91-8.22), 0.069	1.29 (0.38-4.34), 0.638	1.03 (0.27-3.94), 0.964	-
		1995-99	**3.16** (1.18-8.44), 0.027	1.28 (0.35-4.65), 0.667	0.99 (0.24-4.00), 0.983	0.94 (0.65-1.37), 0.718
		2000-04	2.19 (0.97-4.95), 0.057	1.44 (0.51-4.10), 0.440	1.02 (0.28-3.77), 0.969	1.15 (0.80-1.67), 0.395
		2005-09	-	-	1.56 (0.38-6.35), 0.502	1.15 (0.78-1.70), 0.420
		2010-14	1.52 (0.63-3.68), 0.302	1.23 (0.40-3.81), 0.687	1.22 (0.32-4.59), 0.751	0.82 (0.52-1.28), 0.332
	Model p-value, residual I^2, adjusted R^2		*0.210, 81.4%, 45.8%*	*0.394, 97.5%, 12.4%*	*0.244, 98.6%, 24.9%*	*0.067, 50.2%, 78.8%*
	HR/rate ratio	1980-84 (ref)	Insufficient observations	Insufficient observations	1	Insufficient observations
		1985-89			-	
		1990-94			-	
		1995-99			2.00 (0.46-8.59), 0.306	
		2000-04			1.16 (0.27-4.95), 0.823	
		2005-09			1.68 (0.42-6.77), 0.417	
		2010-14			-	
	Model p-value, residual I^2, adjusted R^2				*0.328, 95.6%, 19.3%*	
Random effects meta-analysis I^2		*SMRs*	*91.5%*	*98.0%*	*99.2%*	*80.4%*
		HR/rate ratio			*98.9%*	

CVA – cerebrovascular accident, CHD – coronary heart disease, CVD – all circulatory disease, SMR – standardised mortality ratio, HR – hazard ratio, ref – reference group. Results where 95% confidence intervals exclude the null highlighted in bold. Models adjusted for age and sex and minimum number of additional factors.



SCZ – schizophrenia, BD – bipolar disorder, CVA – cerebrovascular accident, CHD – coronary heart disease, CVD – all circulatory disease,
SMR – standardised mortality ratio, HR – hazard ratio, RRatio – rate ratio
y-axes show estimate of effect size on log scale, shaded areas show 95% confidence intervals for risk estimates

Fig C: Trend in risk of CVD mortality for severe mental illness compared with controls, by median 5-year calendar period of outcome



SCZ – schizophrenia, BD – bipolar disorder, CVA – cerebrovascular accident, CHD – coronary heart disease, CVD – major cardiovascular events, HR – hazard ratio, RRatio – rate ratio, OR – odds ratio, RR – risk ratio
y-axes show estimates of effect size on log scale, shaded areas show 95% confidence intervals for risk estimates

Fig D: Trend in risk of CVD incidence for severe mental illness compared with controls, by median 5-year calendar period of outcome